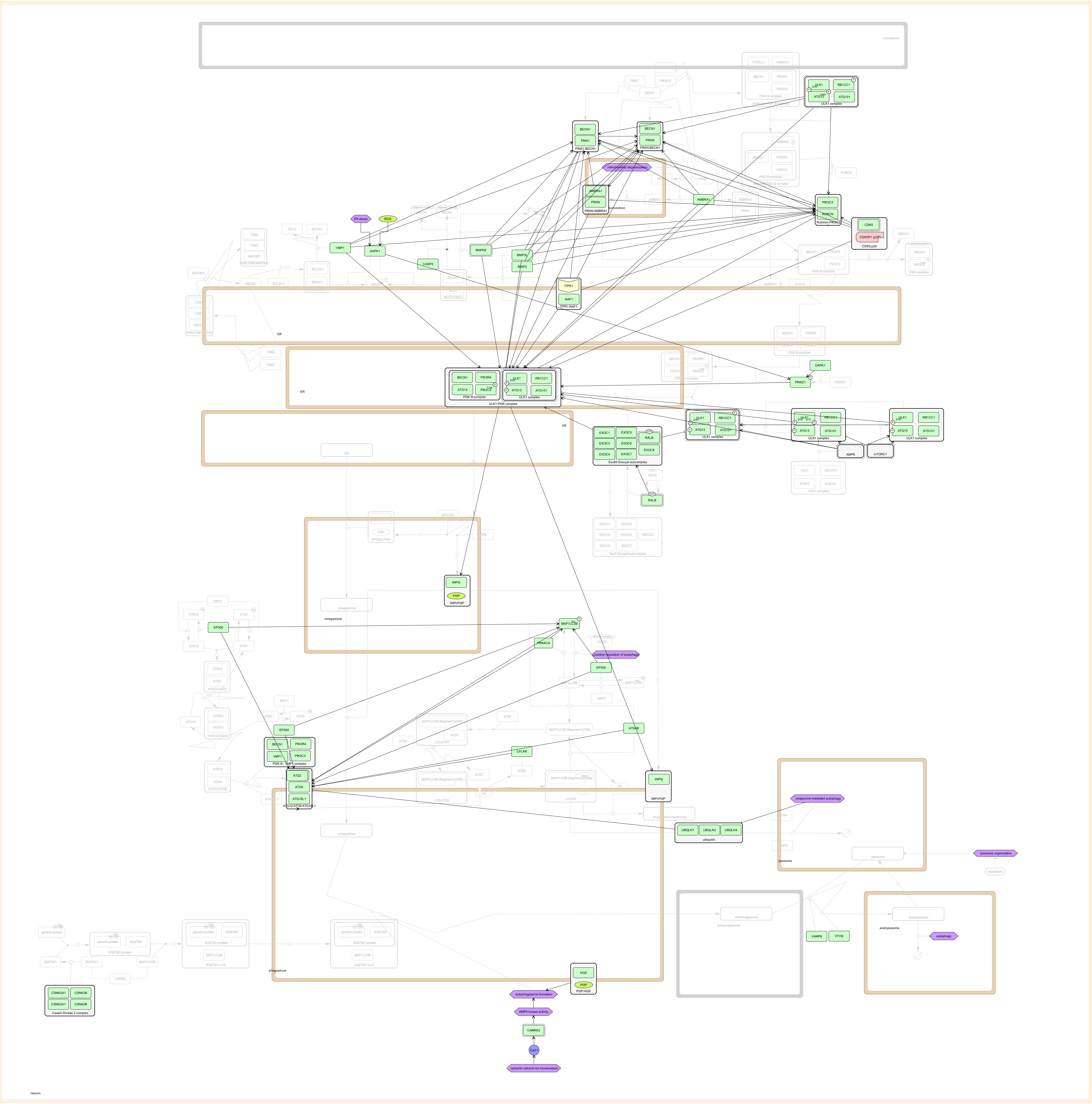
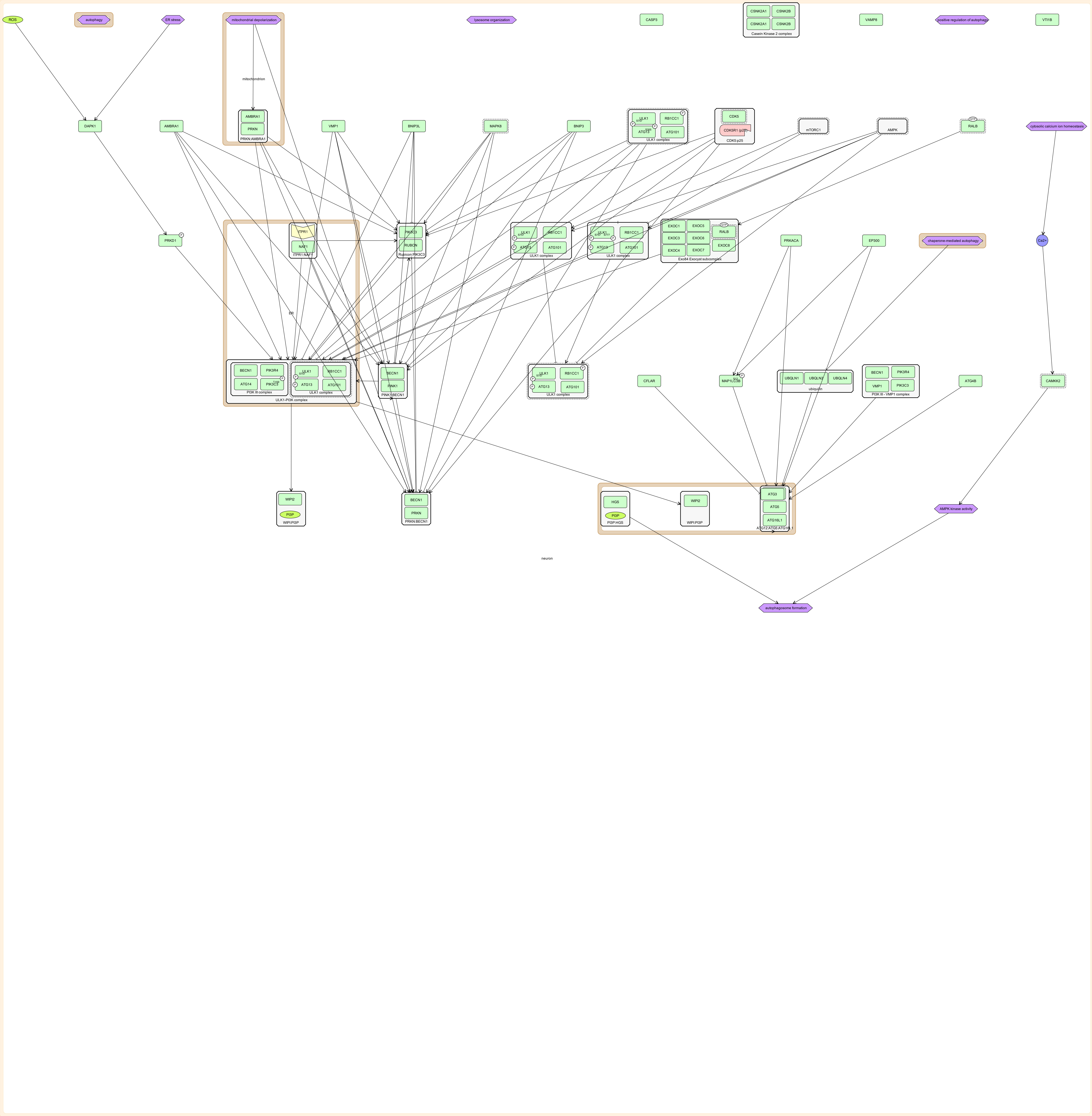

microtubule
DYNLL1
AMBRA1
BECN1
PIK3R4
PIK3C3
PI3K III complex
anchored PI3K III complex
PINK1
ULK1
S757
RB1CC1
ATG13
S355
ATG101
ULK1 complex
BECN1
PINK1
PINK1:BECN1
BECN1
PRKN
PRKN:BECN1
AMBRA1
PI3K III complex
released PI3K III complex
RUBCN
mitochondrial depolarization
PRKN
AMBRA1
PRKN
PRKN:AMBRA1
mitochondrion
AMBRA1
AMBRA1
PRKN
PRKN:AMBRA1
PIK3C3
RUBCN
Rubicon:PIK3C3
CDK5
CDK5R1 (p25)
CDK5:p25
BECN1
ER stress
ROS
BCL2
BECN1
TAB2
TAB3
MAP3K7
TAB2:TAB3:MAP3K7
VMP1
DAPK1
MAPK8
BNIP3L
BNIP3
CASP3
BCL2L1
BECN1
BCL2L1
BCL2L1:BECN1
MAP3K7
BECN1
BCL2L1
BECN1
BECN1:BCL2
BCL2
ITPR1
NAF1
ITPR1:NAF1
PI3K III complex
PIK3R4
PIK3C3
PIK3 complex
phosphorylated
T159
AMBRA1
ATG14
TAB2
TAB3
BECN1
TAB2:TAB3:BECN1
ER
TAB3
TAB2
BECN1
PIK3R4
ATG14
PIK3C3
PI3K III complex
DAPK1
PRKD1
PRKD1
ER
BECN1
PIK3R4
ATG14
PIK3C3
T159
PI3K III complex
ULK1
S757
RB1CC1
ATG13
ATG101
ULK1 complex
ULK1-PI3K complex
ER
ER
EXOC1
EXOC5
EXOC3
EXOC6
EXOC4
EXOC7
RALB
GTP
EXOC8
Exo84 Exocyst subcomplex
ULK1
RB1CC1
ATG13
ATG101
ULK1 complex
ULK1
S757
S777
RB1CC1
ATG13
ATG101
ULK1 complex
ULK1
RB1CC1
ATG13
ATG101
ULK1 complex
AMPK
mTORC1
ULK1
RB1CC1
ATG13
ATG101
ULK1 complex
RALB
GDP
RALB
GTP
EXOC1
EXOC5
EXOC3
EXOC6
EXOC2
EXOC4
EXOC7
Sec5 Exocyst subcomplex
ZFYVE1
PI3P
ZFYVE1:PI3P
WIPI2
WIPI2
PI3P
WIPI:PI3P
omegasome
omegasome
SIRT1
ATG12
ATG7
EP300
ATG12
ATG7
MAP1LC3B
S12
PRKACA
LRRK2
G2019S mutation
positive regulation of autophagy
EP300
ATG12
ATG7
ATG12:ATG7
MAP1LC3B
MAP1LC3B
SIRT1
SIRT1
ATG10
ATG12
ATG10
ATG12:ATG10
ATG5
ATG5
EP300
BECN1
PIK3R4
VMP1
PIK3C3
PI3K III - VMP1 complex
MAP1LC3B (fragment 2-120)
ATG7
LC3:ATG7
ATG7
MAP1LC3B (fragment 2-120)
ATG3
ATG4B
CFLAR
ATG3
ATG7
ATG12
ATG5
ATG12:ATG5
ATG3
ATG5
ATG16L1
ATG12:ATG5:ATG16L1
ATG16L1
MAP1LC3B (fragment 2-120)
LC3:ATG3
PE
MAP1LC3B (fragment 2-120)
PI3P
PE
LC3:PE
PE
WIPI2
WIPI:PI3P
phagophore membrane
phagophore
UBQLN1
UBQLN2
UBQLN4
ubiquilin
chaperone-mediated autophagy
LAMP2A
HSPA4
LAMP3
lysosome
lysosome
lysosome organization
rapamycin
generic protein
Ub
generic protein
SQSTM1
SQSTM1:protein
generic protein
SQSTM1
SQSTM1:protein
MAP1LC3B
SQSTM1:LC3
generic protein
SQSTM1
SQSTM1:protein
MAP1LC3B
SQSTM1:LC3
SQSTM1
SQSTM1
MAP1LC3B
LRRK2
autophagosome
autophagosome
VAMP8
VTI1B
autolysosome
autolysosome
autophagy
phagophore
HGS
PI3P
PI3P:HGS
autophagosome formation
AMPK kinase activity
CAMKK2
Ca2+
cytosolic calcium ion homeostasis
CSNK2A1
CSNK2B
CSNK2A1
CSNK2B
Casein Kinase 2 complex
neuron


ROS
autophagy
ER stress
mitochondrial depolarization
mitochondrion
lysosome organization
CASP3
CSNK2A1
CSNK2B
CSNK2A1
CSNK2B
Casein Kinase 2 complex
VAMP8
positive regulation of autophagy
VTI1B
DAPK1
AMBRA1
AMBRA1
PRKN
PRKN:AMBRA1
VMP1
BNIP3L
MAPK8
BNIP3
ULK1
S757
P
RB1CC1
P
ATG13
S355
P
ATG101
ULK1 complex
CDK5
CDK5R1 (p25)
CDK5:p25
mTORC1
AMPK
GTP
RALB
cytosolic calcium ion homeostasis
PRKD1
P
ITPR1
NAF1
ITPR1:NAF1
PIK3C3
RUBCN
Rubicon:PIK3C3
ULK1
S757
P
RB1CC1
P
ATG13
ATG101
ULK1 complex
ULK1
S757
S777
P
P
RB1CC1
P
ATG13
ATG101
ULK1 complex
EXOC1
EXOC5
GTP
RALB
EXOC3
EXOC6
EXOC8
EXOC4
EXOC7
Exo84 Exocyst subcomplex
PRKACA
EP300
chaperone-mediated autophagy
Ca2+
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BECN1
PIK3R4
ATG14
PIK3C3
T159
P
PI3K III complex
ULK1
S757
P
RB1CC1
P
ATG13
ATG101
ULK1 complex
ULK1-PI3K complex
BECN1
PINK1
PINK1:BECN1
ULK1
S757
P
RB1CC1
P
P
ATG13
ATG101
ULK1 complex
CFLAR
MAP1LC3B
S12
P
UBQLN1
UBQLN2
UBQLN4
ubiquilin
BECN1
PIK3R4
VMP1
PIK3C3
PI3K III - VMP1 complex
ATG4B
CAMKK2
WIPI2
PI3P
WIPI:PI3P
BECN1
PRKN
PRKN:BECN1
HGS
PI3P
PI3P:HGS
WIPI2
WIPI:PI3P
ATG3
ATG5
ATG16L1
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AMPK kinase activity
neuron
autophagosome formation